

CLAREMORE, OK, USA

WMO: 722091

Lat: 36.294N

Lon: 95.479W

Elev: 221

StdP: 98.70

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 53940

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.1	-7.9	-17.1	0.9	-9.8	-13.9	1.2	-6.4	11.2	12.3	10.5	12.4	2.9	20	0.473

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.7	37.1	23.7	35.0	23.9	32.8	23.9	25.9	32.1	25.4	31.7	24.8	31.0	4.0	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.1	19.5	29.2	23.0	18.3	28.2	22.7	17.9	27.9	81.9	32.3	78.6	32.0	75.9	30.7	28.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.5	8.9	8.0	-15.5	38.3	4.1	2.8	-18.5	40.4	-20.9	42.0	-23.1	43.6	-26.1	45.7		
(5)			-16.0	26.7	3.8	1.0	-18.7	27.4	-20.9	28.0	-23.1	28.5	-25.8	29.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	15.2	3.0	4.7	10.6	15.0	19.7	24.6	26.6	26.1	22.2	15.6	9.7	4.1		
	DBStd	9.63	5.91	6.36	5.79	4.87	4.32	2.70	2.84	3.18	3.76	5.03	5.26	5.47		
	HDD10.0	746	228	169	63	13	1	0	0	0	0	10	68	194		
	HDD18.3	2091	476	382	246	121	38	1	0	0	11	114	262	441		
	CDD10.0	2649	10	21	82	162	301	438	516	499	366	184	58	12		
	CDD18.3	953	0	1	7	20	79	189	257	241	126	30	3	0		
	CDH23.3	9618	1	6	48	174	636	1878	2897	2539	1154	263	22	0		
	CDH26.7	4192	0	1	7	35	166	761	1458	1249	453	62	2	0		
(14) Wind	WSAvg	3.4	3.5	3.8	4.2	4.2	3.5	3.2	2.5	2.8	2.7	3.1	3.6	3.4		
(15) Precipitation	PrecAvg	1132	56	53	87	118	134	133	93	85	121	92	94	64		
	PrecMax	1385	114	133	214	231	237	271	219	227	288	206	216	147		
	PrecMin	781	2	6	20	49	58	40	29	2	28	26	6	2		
	PrecStd	157	31	37	57	55	56	64	56	60	67	50	62	42		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	21.3	23.8	27.5	31.1	32.0	36.1	39.1	41.1	35.2	31.1	26.2	21.4		
		MCWB	13.5	15.5	17.8	19.9	22.2	24.3	23.3	23.8	23.1	22.1	17.2	16.6		
	2%	DB	17.9	21.0	24.2	27.3	30.1	33.0	37.3	37.2	32.7	28.0	22.7	17.6		
		MCWB	12.2	14.1	16.5	18.4	22.1	24.1	23.7	23.6	23.3	19.8	15.6	12.5		
	5%	DB	15.9	17.7	22.3	25.2	28.0	32.3	35.1	34.9	31.3	26.4	20.9	15.2		
		MCWB	10.4	12.5	15.1	17.1	21.4	24.0	23.7	24.1	22.8	18.4	15.2	10.8		
	10%	DB	12.5	15.2	20.0	22.9	27.2	31.1	32.9	32.6	29.2	24.0	17.8	12.7		
		MCWB	8.0	10.4	13.8	16.5	20.9	23.7	24.3	23.7	21.9	18.0	13.1	9.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	16.2	17.5	19.3	21.3	24.4	26.3	27.6	26.6	25.1	23.2	19.4	17.3		
		MCDB	18.7	21.2	25.7	27.5	29.2	32.8	32.4	33.1	31.6	29.0	23.6	19.5		
	2%	WB	13.6	15.6	17.7	20.0	23.4	25.6	26.2	25.7	24.3	22.0	17.6	14.5		
		MCDB	16.9	19.3	22.9	25.1	28.2	31.6	31.7	32.4	30.5	26.8	21.2	17.0		
	5%	WB	10.8	13.6	16.4	19.0	22.5	24.8	25.5	25.0	23.6	20.7	15.8	12.1		
		MCDB	15.0	17.2	20.8	23.5	27.3	30.4	31.7	31.7	29.4	23.5	19.1	13.5		
	10%	WB	8.1	10.7	15.1	17.9	21.6	24.0	24.9	24.3	22.8	19.3	14.3	9.8		
		MCDB	11.8	14.7	18.8	21.8	26.0	29.4	31.2	31.0	28.1	22.9	17.8	12.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	11.5	11.5	11.9	12.1	10.9	10.7	11.7	11.4	12.0	12.6	12.3	10.8		
		MCDBR	16.0	15.6	14.3	14.2	12.0	12.2	14.2	14.2	13.3	14.3	14.7	14.2		
	5% WB	MCWBR	10.5	9.8	8.1	7.8	5.9	4.9	4.2	4.4	5.3	7.1	9.0	9.2		
		MCWBR	14.0	13.8	11.7	11.7	10.4	10.5	11.0	11.4	11.3	10.3	12.1	11.1		
(40) Clear-Sky Solar Irradiance	taub		0.303	0.310	0.355	0.386	0.427	0.430	0.444	0.446	0.394	0.350	0.316	0.297		
		taud	2.478	2.475	2.384	2.315	2.261	2.281	2.285	2.250	2.363	2.470	2.509	2.512		
	Ebn at Noon		895	932	912	896	861	855	839	830	859	874	874	881		
		Edh at Noon	83	93	112	127	136	134	132	134	114	93	80	75		
(44) All-Sky Solar Radiation	RadAvg		2.51	3.10	4.09	5.13	5.69	6.52	6.55	5.93	4.96	3.69	2.69	2.13		
	RadStd		0.24	0.46	0.23	0.31	0.53	0.62	0.40	0.40	0.39	0.46	0.32	0.26		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (33 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page